Fixed-m=2 Semaev: an unbounded-LFD proof search

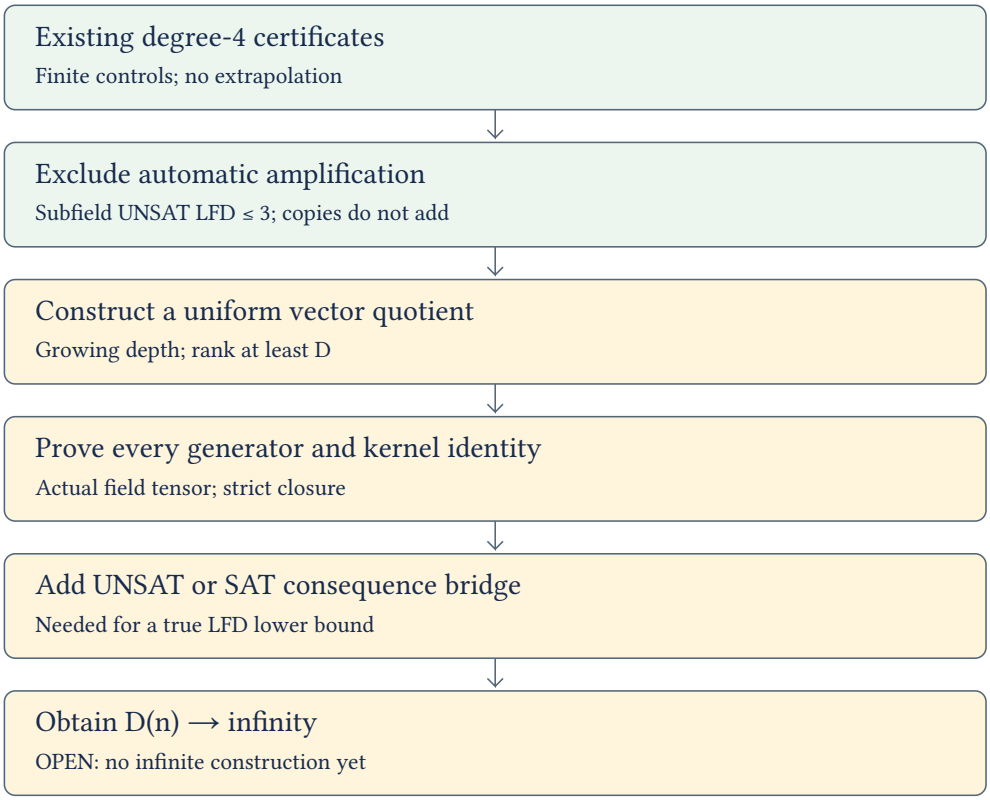
2026-11-09 — written proof research. No new solver, build, cloud job, or certificate replay.

Target: normalized multilinear binary S3 descent, Boolean field equations, dimension k=ceil(n/2). Construct infinitely many instances with last fall degree tending to infinity. **The target remains open.**

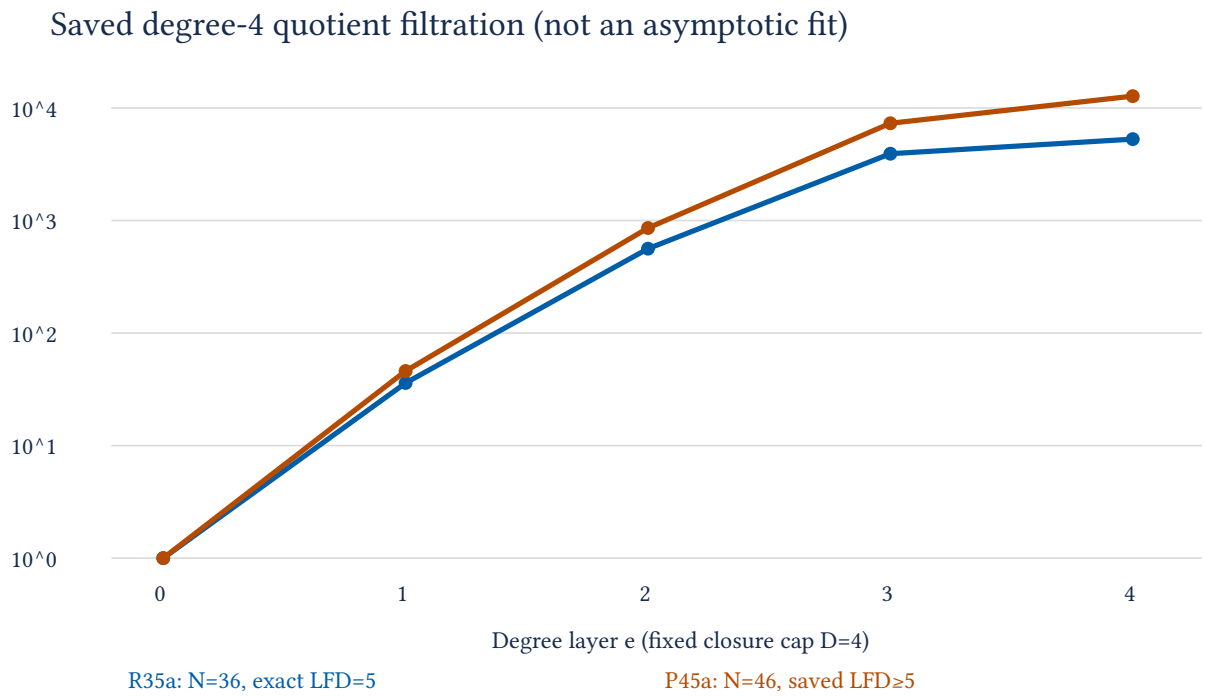
Two derived proof results

Subfield obstruction. If n=2k and V is the quadratic-extension subfield, every unsatisfiable instance has LFD at most 3. Splitting the equations over the subfield yields a linearized constraint on s=X+Y with a one-dimensional kernel. Fixing its remaining bit leaves affine systems; one-bit branching lifts their contradictions at degree 3. Two independent approaches checked the argument. It is not a bound on arbitrary subspaces or on raw quartic presentations.

Certificate filtration obstruction. For a closed-kernel certificate at cap D, an image-filtration plateau before layer D-1 gives commuting idempotent coordinate actions and hence a Boolean solution. Therefore an UNSAT certificate must grow strictly through layer D-1 and have rank at least D. This necessary condition does not construct the desired family.



Existing finite certificates as controls



Exact arithmetic from saved closure profiles, one named instance per curve. Vertical scale is base-10 logarithmic quotient dimension. No statistical uncertainty or asymptotic fit is inferred. R35a has independently reproduced exact LFD=5 despite quotient dimension 5287 at degree 4. P45a's cited profile is implementation-A evidence.

Remaining construction

Specify varying half-dimensional subspaces and quotient maps at arbitrarily growing degree caps. Prove generator annihilation and closure of the entire low-degree kernel under each variable, then prove UNSAT or a missing SAT ideal consequence. Respect the actual field multiplication tensor and target admissibility. Copying an old example, extending its ambient field with V fixed, or displaying a high-degree eliminant does not establish growth.

Sources and review scope

Full arguments and limitations: report.md, algebraic-route.md, certificate-route.md and definition-audit.md. Source hashes: source-manifest.json. Prior open obligation: problems/growing-boolean-semaev-lower-bound.md at dadfb1b2a5b6e8e6045f8547d26ec2b40d4ff81a. Historical evidence: analysis/semaev-m2-last-fall-degree/results/verification.md.

Primary references: Caminata–Gorla, arXiv:2112.05579, Section 3; Huang–Kosters–Yang–Yeo, arXiv:1505.02532; Kosters–Yeo, arXiv:1503.08001. The prime-digit family proof in the repository does not imply the characteristic-two result. These are written proofs and source review, not Lean certification or a new finite computational result.